

Anomaly Detection & Dimensionality Reduction

Isolation Forest, t-SNE, UMAP & PCA

Two independent anomaly detection projects with advanced dimensionality reduction visualization

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Project 1 — Jet Engine Anomaly Detection

Detect sensor anomalies in jet engine telemetry using Isolation Forest with t-SNE/UMAP visualization

Overview

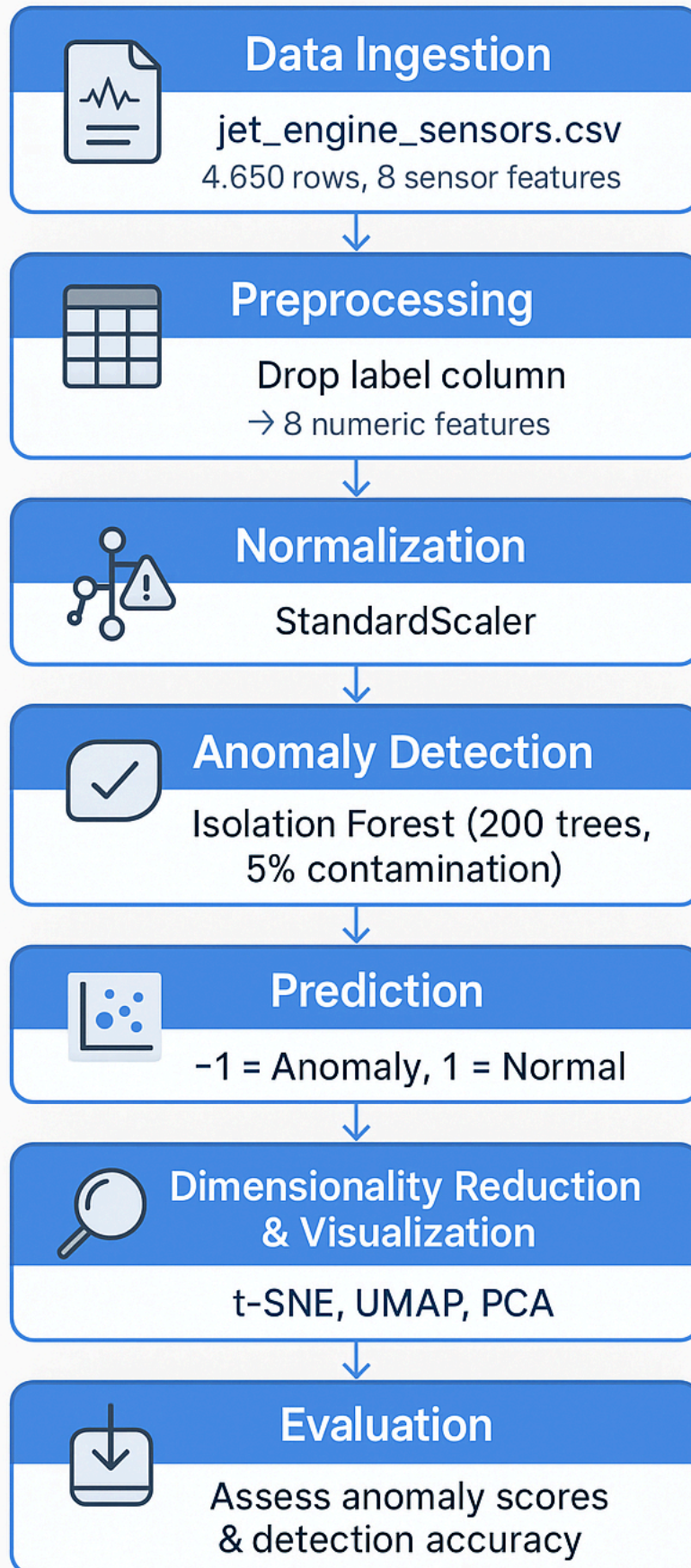
Detect anomalous sensor readings from jet engine telemetry data. The dataset contains 4,650 recordings from 8 sensors with a 3.2% anomaly rate. Isolation Forest identifies anomalies by their isolation depth in random trees, while t-SNE and UMAP provide powerful 2D projections for visual analysis.

Dataset A: Jet Engine Sensors

Property	Value
Rows	4,650 (4,500 normal + 150 anomalies)
Features	turbine_temp_c, oil_pressure_psi, vibration_hz, fuel_flow_rate, exhaust_gas_temp_c, fan_speed_rpm, compressor_ratio, n1_speed_pct
Anomaly Rate	3.2%
Anomaly Types	Turbine overheat, Oil pressure loss, Fan blade imbalance
Task	Unsupervised anomaly detection

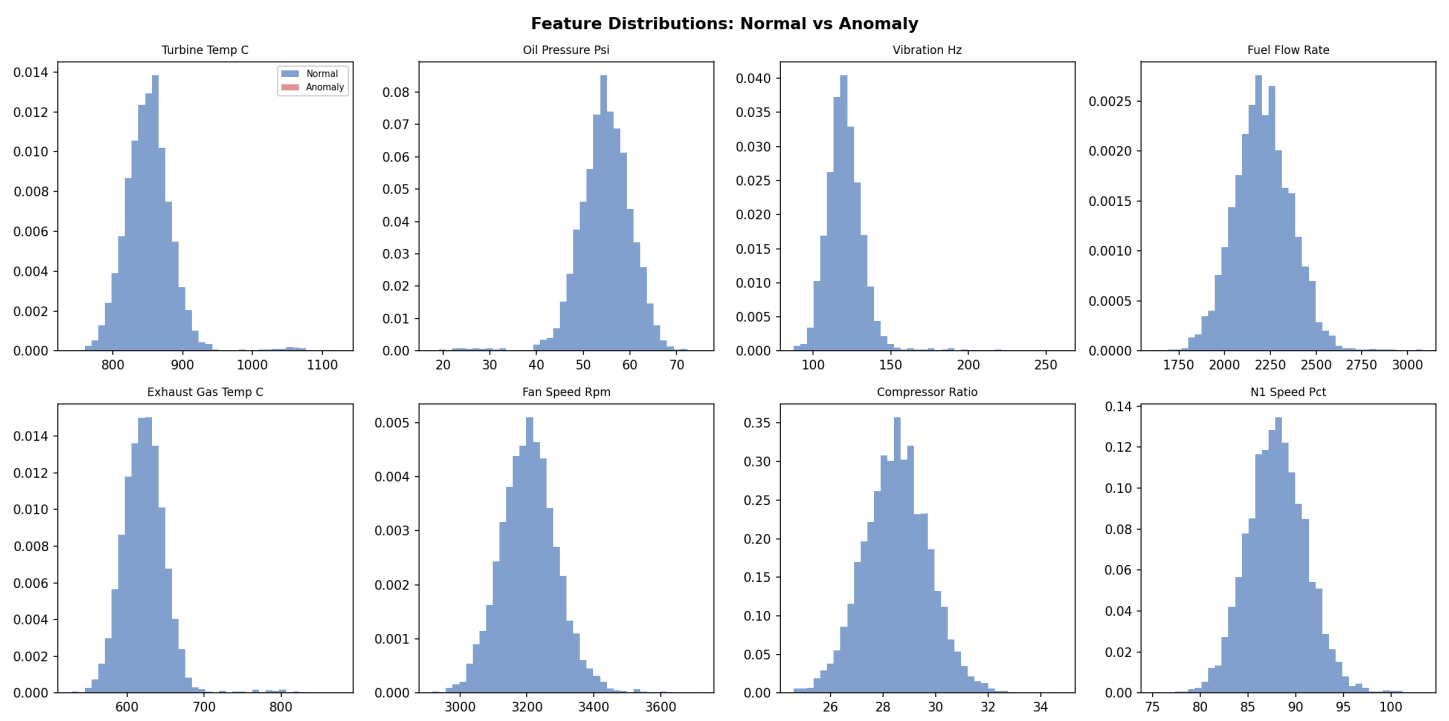
Flowchart

Jet Engine Anomaly Detection Pipeline



Exploratory Data Analysis

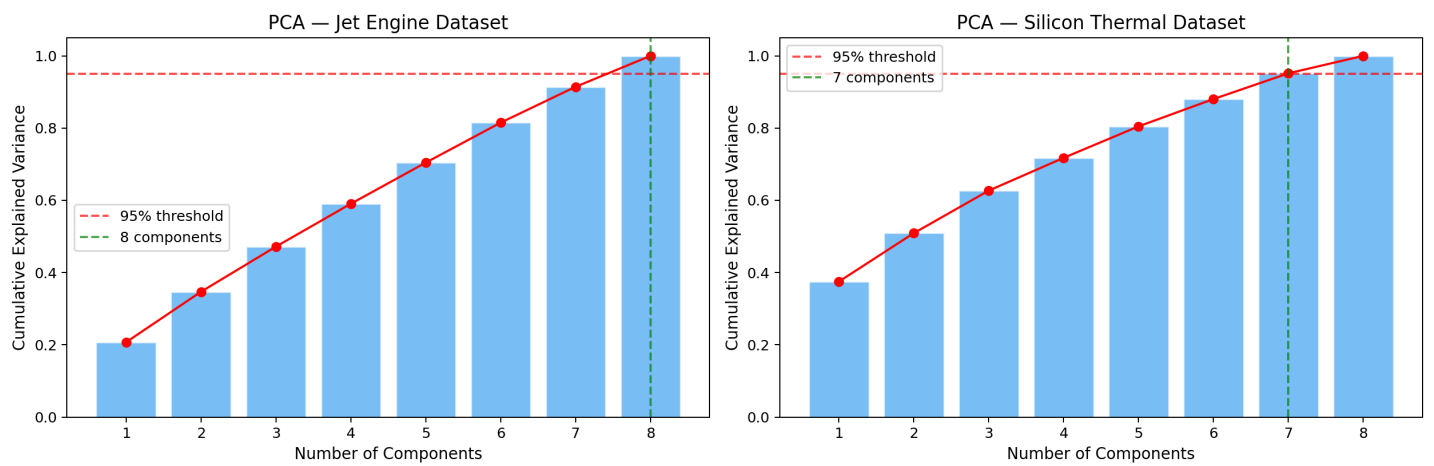
Feature distributions and correlation matrix for jet engine sensor readings.



PCA Explained Variance

Cumulative explained variance showing how many principal components are needed to capture the data distribution.

Dimensionality Reduction Analysis

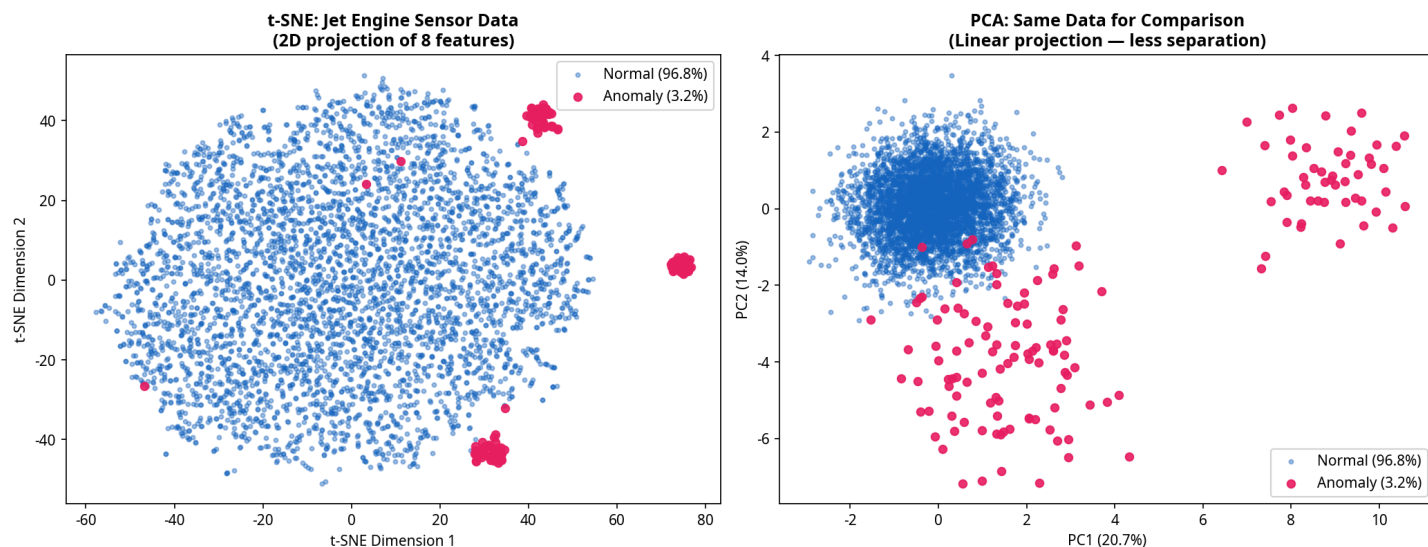


t-SNE vs PCA Comparison

Side-by-side comparison of t-SNE (nonlinear) vs PCA (linear) projections — t-SNE reveals cluster structure that PCA misses.

t-SNE vs PCA: Non-linear vs Linear Dimensionality Reduction

t-SNE reveals cluster structure invisible to PCA

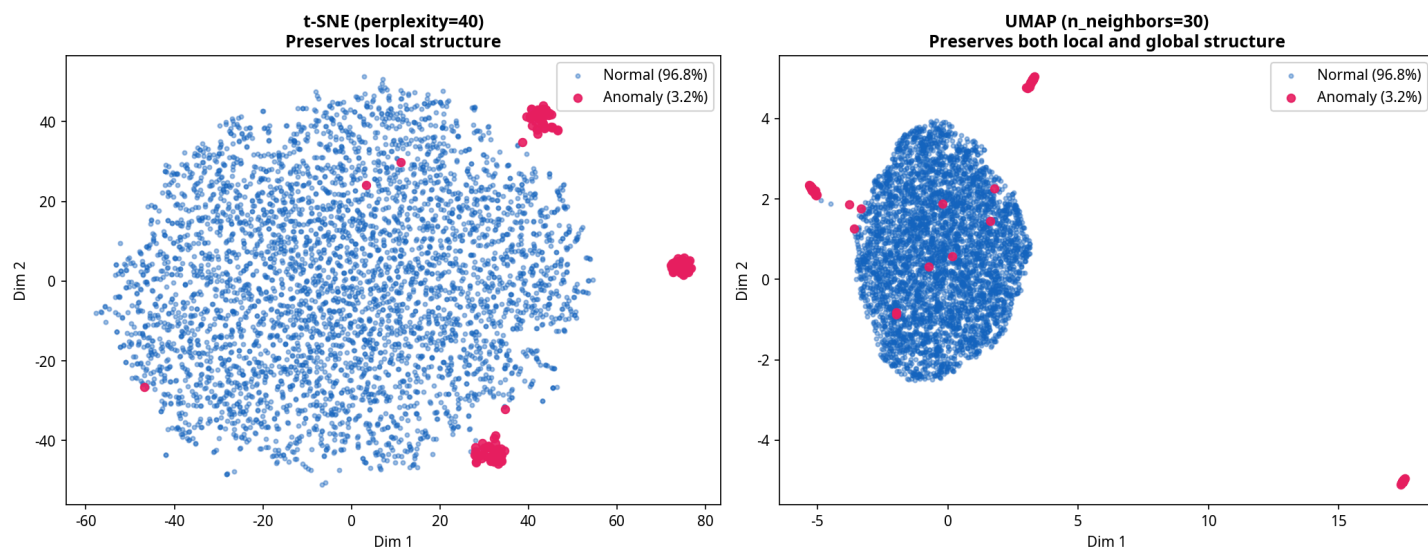


UMAP vs t-SNE Comparison

UMAP preserves more global structure than t-SNE while maintaining local neighborhoods.

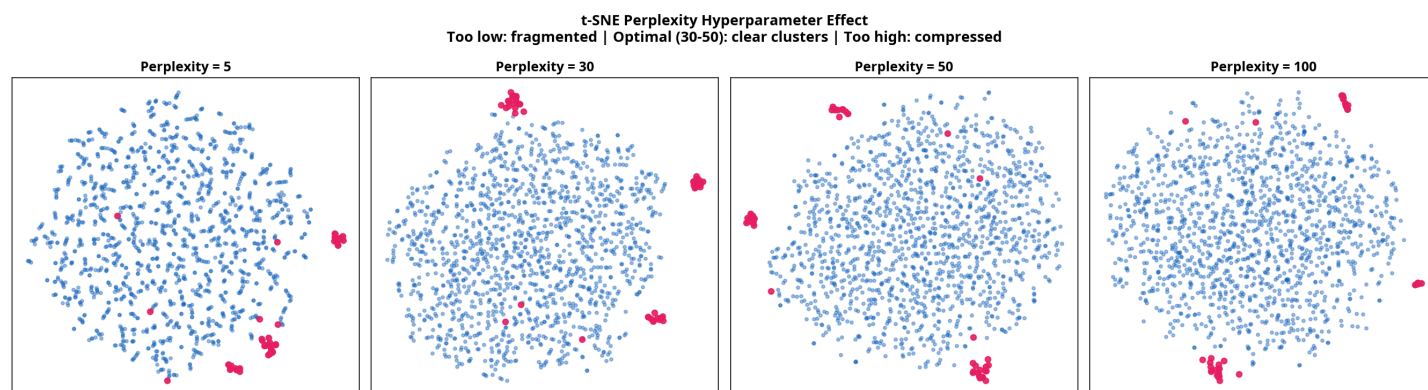
t-SNE vs UMAP: Non-linear Dimensionality Reduction Comparison

Both reveal hidden structure; UMAP is faster and preserves global topology



t-SNE Perplexity Effects

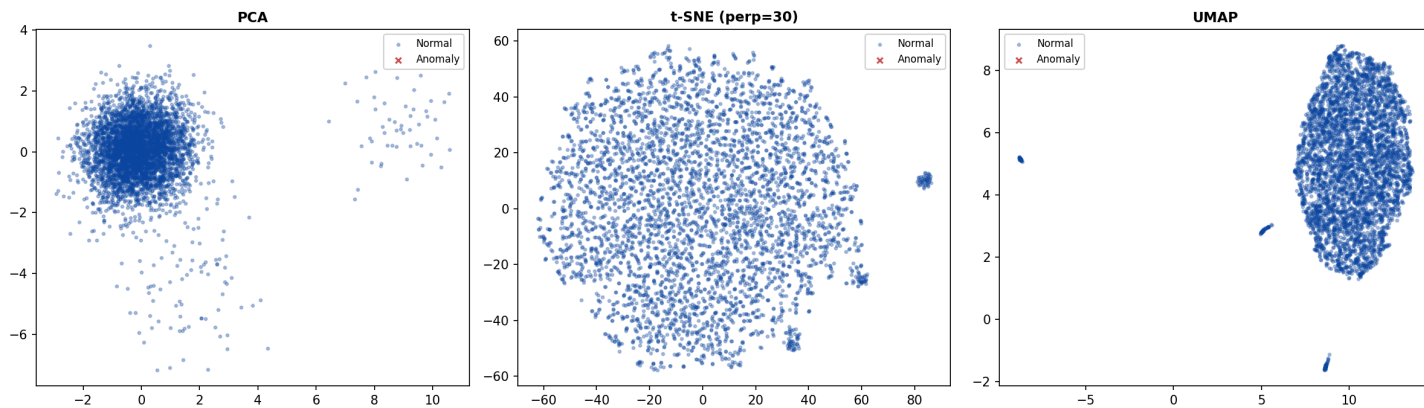
How perplexity (5, 15, 30, 50) affects the t-SNE embedding — controls the balance between local and global structure.



Dimensionality Reduction Comparison

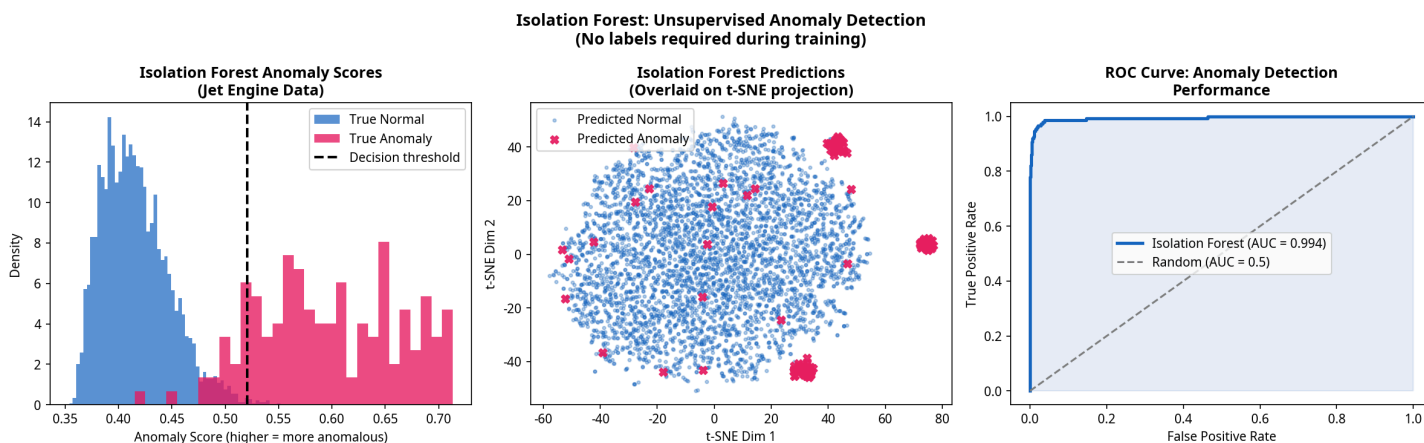
Multi-method comparison: PCA, t-SNE, UMAP side-by-side on jet engine data.

Dimensionality Reduction Comparison: Anomaly Separation



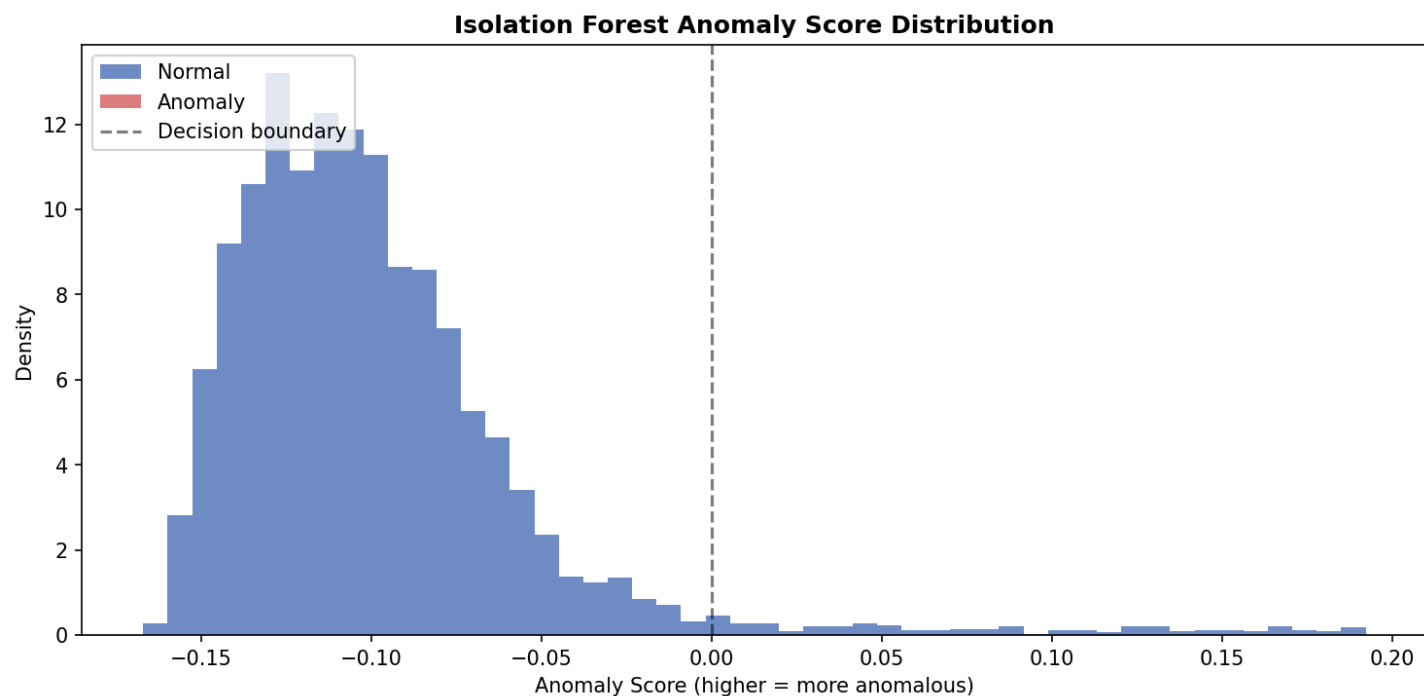
Isolation Forest Results

Isolation Forest anomaly detection with contamination=0.05. Normal points vs detected anomalies.



Anomaly Score Distribution

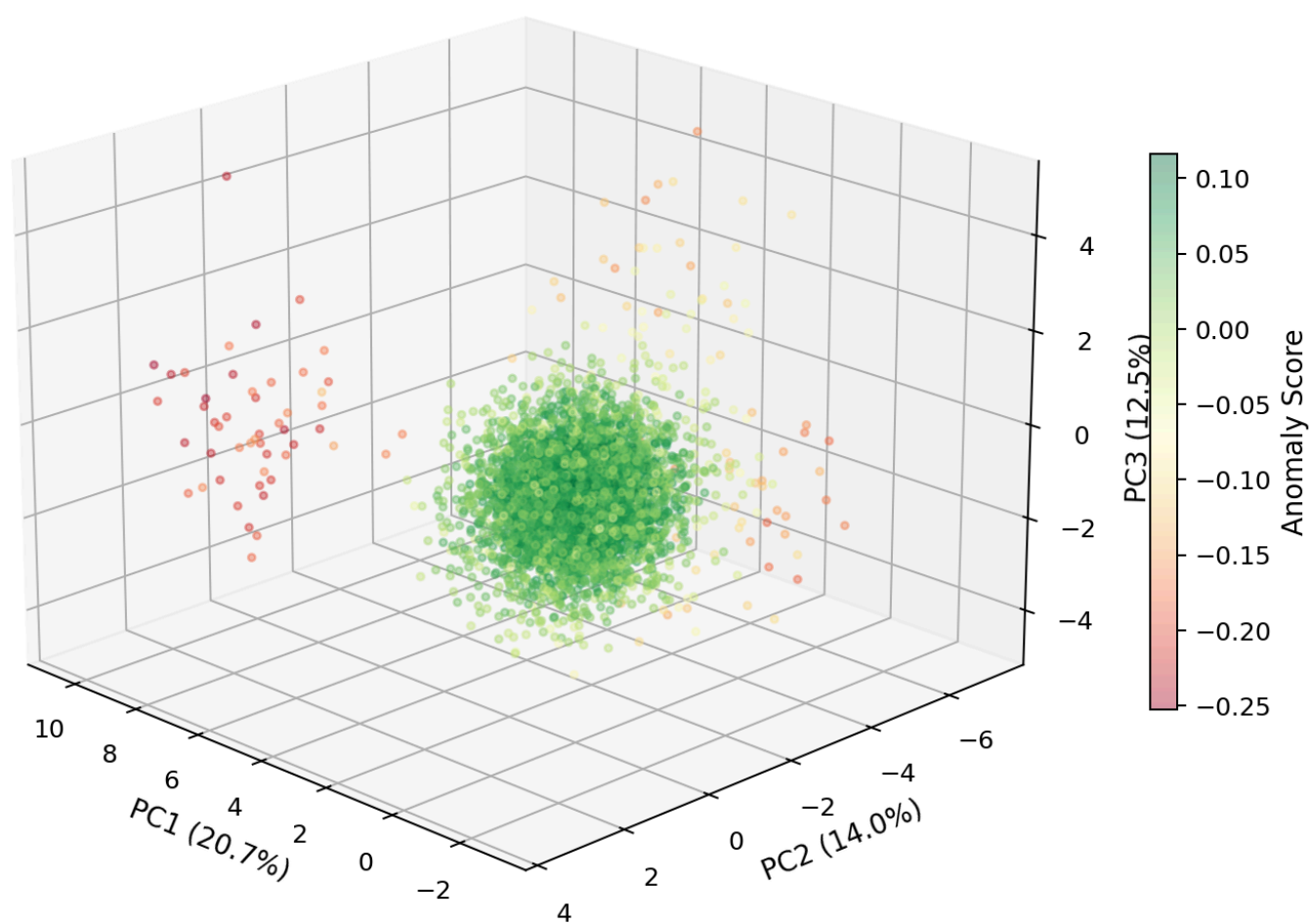
Distribution of Isolation Forest anomaly scores — lower scores indicate more anomalous points.



3D Anomaly Visualization

Three-dimensional projection of jet engine sensor space with anomalies highlighted.

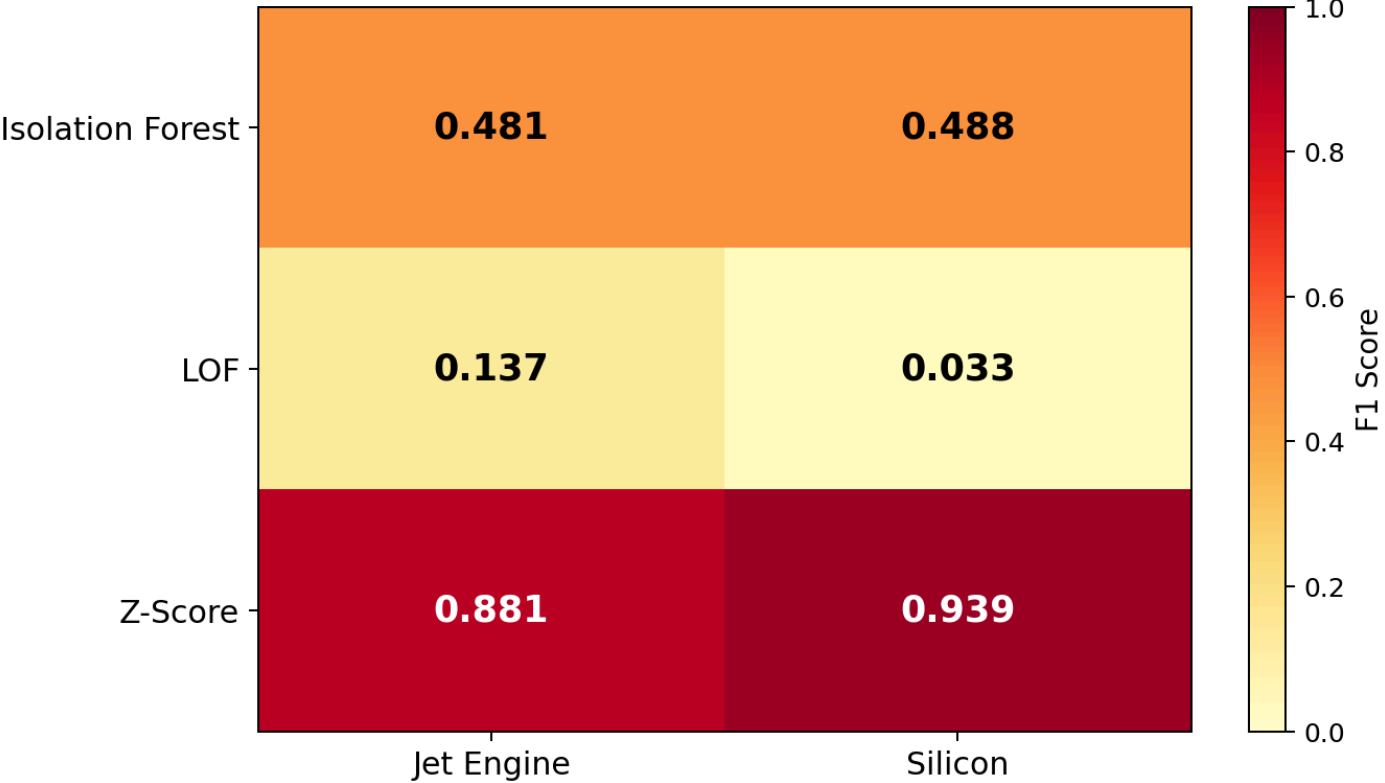
Isolation Forest Anomaly Scores in PCA Space
Jet Engine Sensor Data



Model Comparison Heatmap

Performance heatmap across anomaly detection methods and parameter settings.

Anomaly Detection Methods Comparison



Production Model Results

Metric	Value
Model	StandardScaler → IsolationForest(n_estimators=200, contamination=0.05)
Anomalies Detected	233 / 4,650 (5.0%)
Estimators	200

Project 2 — Silicon Thermal Hotspot Detection

Identify thermal anomalies in semiconductor chips using Isolation Forest with PCA/t-SNE visualization

Overview

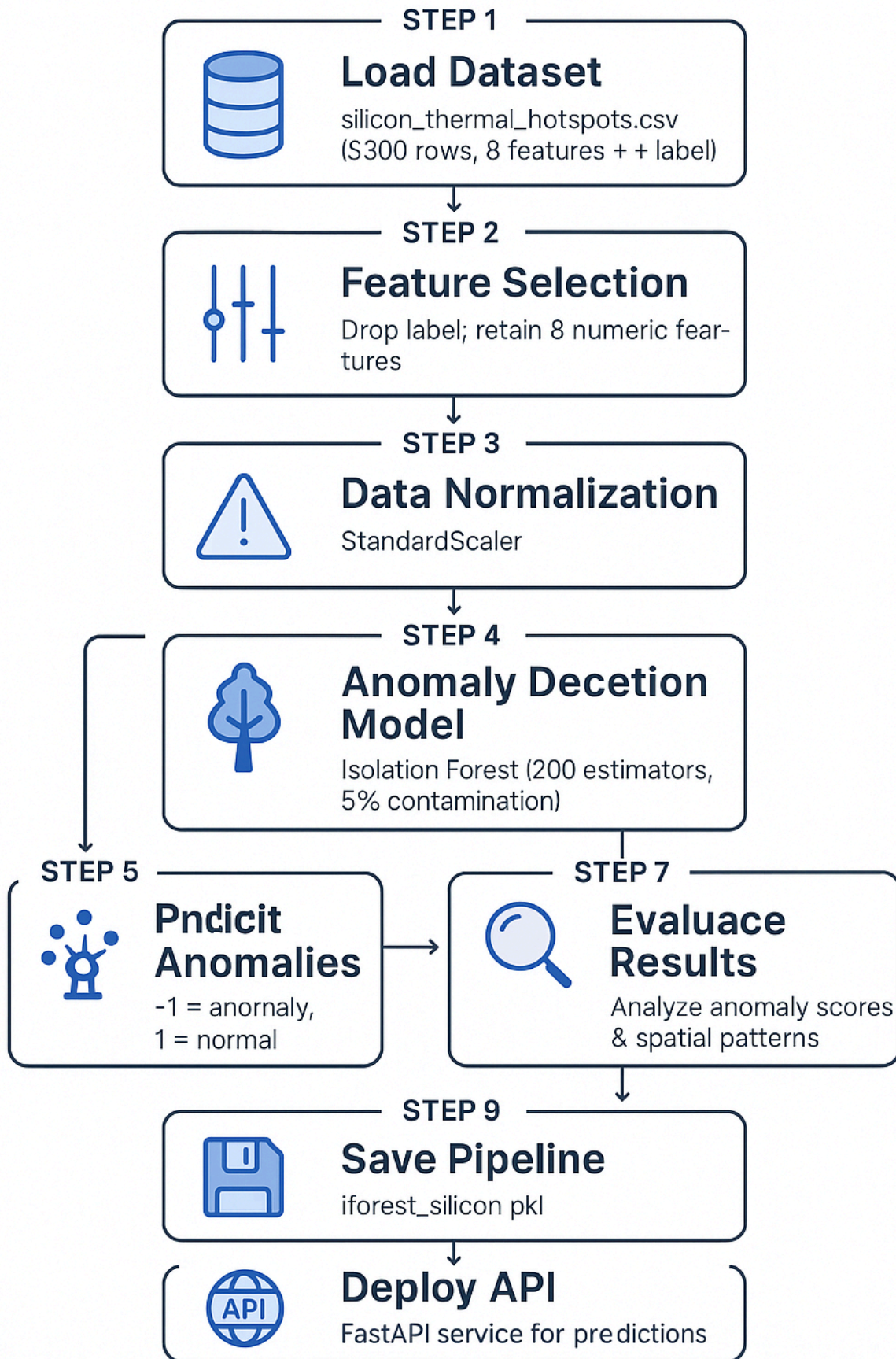
Detect thermal hotspots and electrical anomalies in semiconductor chips from 8 thermal/electrical measurements. The dataset contains 9,300 inspection records with a 3.2% anomaly rate covering thermal runaway, IR drop hotspots, and leakage spikes.

Dataset B: Silicon Thermal Hotspots

Property	Value
Rows	9,300 (9,000 normal + 300 anomalies)
Features	core_temp_c, power_density_mw_mm2, thermal_resistance, leakage_current_ua, clock_freq_ghz, vdd_mv, ir_drop_mv, thermal_gradient
Anomaly Rate	3.2%
Anomaly Types	Thermal runaway, IR drop hotspot, Leakage spike
Task	Unsupervised anomaly detection

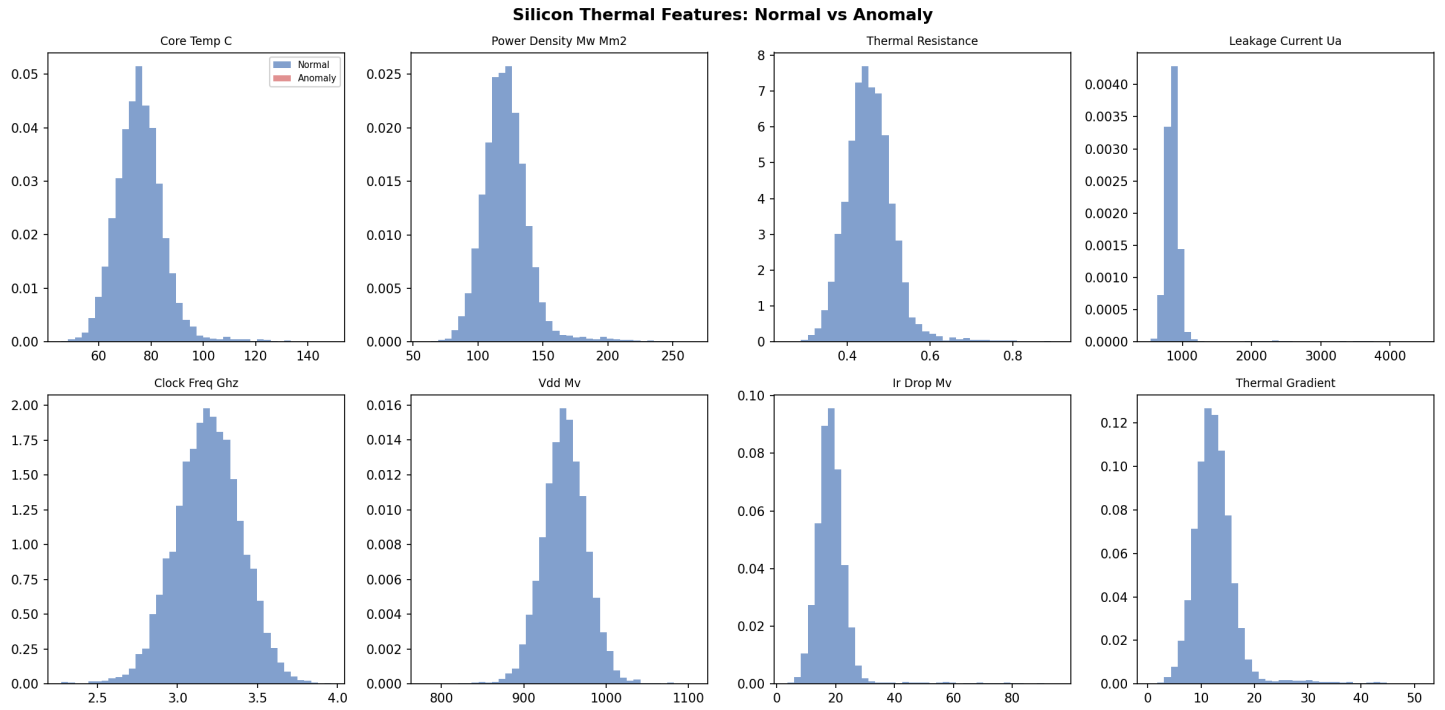
Flowchart

SILICON THERMAL HOTSPOT DETECTION PIPELINE



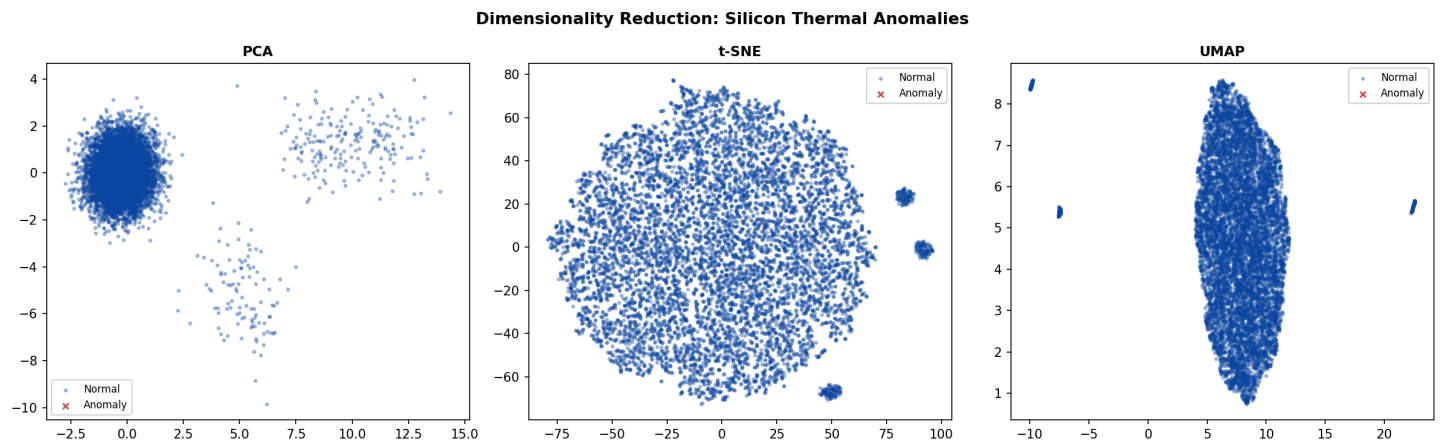
Exploratory Data Analysis

Feature distributions and thermal/electrical correlations for silicon chip data.



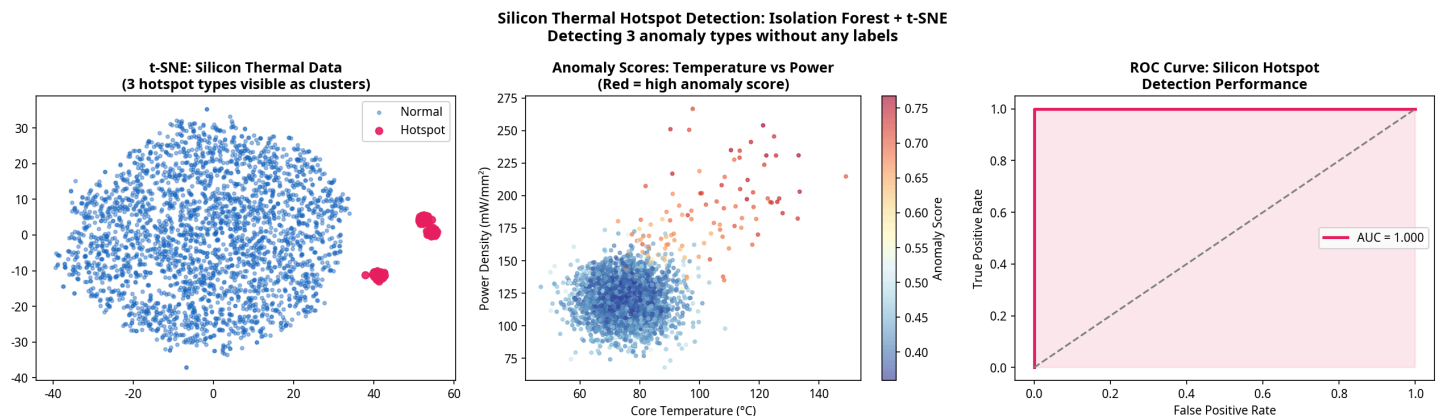
Dimensionality Reduction (t-SNE/PCA)

t-SNE and PCA projections of the silicon thermal feature space, colored by anomaly status.



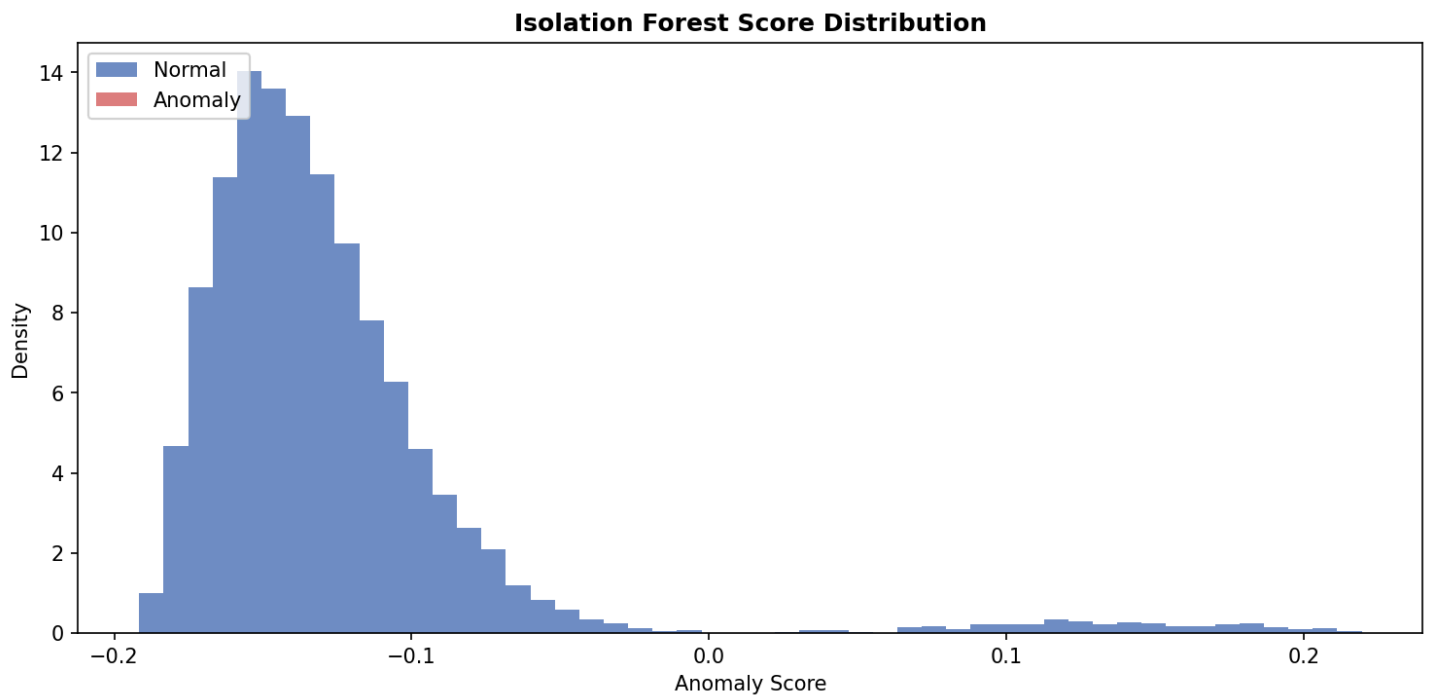
Silicon Hotspot Visualization

Spatial visualization of thermal hotspot locations and anomaly clusters on the chip.



Anomaly Score Distribution

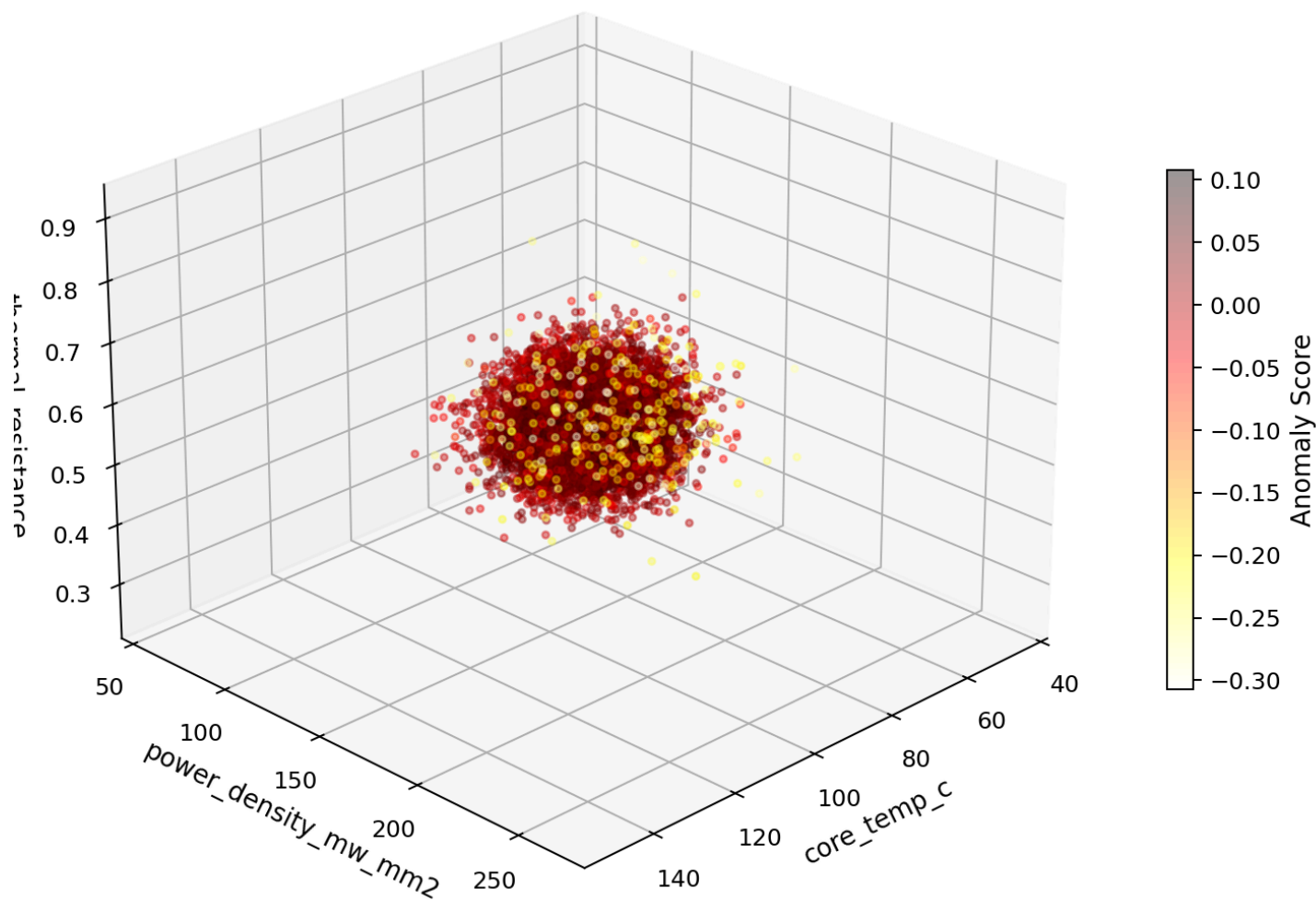
Distribution of Isolation Forest anomaly scores for silicon thermal data.



3D Thermal Visualization

Three-dimensional projection of thermal features with anomalies highlighted in the spatial context.

Silicon Thermal Hotspot Detection
Isolation Forest Scoring



Production Model Results

Metric	Value
Model	StandardScaler → IsolationForest(n_estimators=200, contamination=0.05)
Anomalies Detected	465 / 9,300 (5.0%)
Estimators	200

Appendix — Tech Stack & Key Learnings

Tech Stack

Tool	Version	Purpose
Python	3.11+	Core runtime
scikit-learn	≥ 1.3	IsolationForest, PCA, StandardScaler, Pipeline
umap-learn	≥ 0.5	UMAP dimensionality reduction
pandas	≥ 2.0	Data manipulation
numpy	≥ 1.24	Numerical operations
matplotlib / seaborn	≥ 3.7 / 0.12	Visualizations
FastAPI	latest	REST API serving
joblib	built-in	Model serialization

Key Learnings

#	Insight
1	Isolation Forest's contamination parameter directly controls the detection threshold — set it based on domain knowledge of expected anomaly rate, not just default 0.05.
2	t-SNE is powerful for 2D visualization but non-deterministic and perplexity-sensitive. UMAP preserves more global structure and runs faster on large datasets.
3	PCA explained variance helps decide how many components to keep — 90% threshold is standard, but anomalies often live in the low-variance components that PCA discards.
4	The 5% contamination matches the training detection rate, but the true anomaly rate is 3.2% — the model over-flags some normal points. Precision-recall tradeoff matters in production.
5	StandardScaler before Isolation Forest is critical: features like turbine_temp_c (800-1000) and compressor_ratio (25-30) have wildly different scales that affect tree splits.